

M500 IN-CAR VIDEO SYSTEM AND VIDEOMANAGER EXPLAINED

Configuring your M500 in-car video system to work with VideoManager

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Contents

[Introduction](#)

[Create a group for use with the M500](#)

[Configure media fields for the M500](#)

[Edit the M500 event category media field](#)

[Optionally add other M500 event tags](#)

[Optionally configure incident fields and rules for the M500](#)

[Optionally create an M500 event category incident field](#)

[Optionally enable and configure automatic incident creation](#)

[Set a in-vehicle administrator password](#)

[Create a vehicle network profile](#)

[Create an M500 device profile](#)

[General tab](#)

[Devices tab](#)

[On-Screen text management](#)

[GPS section](#)

[In-Car officer permissions](#)

[Power behavior](#)

[Other](#)

[Recording tab](#)

[Camera](#)

[Watermarks](#)

[Front camera, Front camera wide, Cabin camera, and Rear camera](#)

[Wired 1, Wired 2 and cabin camera microphone](#)

[Recording reminder alerts](#)

[Additional record time](#)

[Auto-start record](#)

[Auto-stop record](#)

[USB groups](#)

[Event tags](#)

[Network tab](#)

[Upload server](#)

[DNS server](#)

[Add your M500 to VideoManager](#)

[Move an M500 to a new vehicle](#)

Introduction

As of VideoManager 16.0.1, it is possible to administer your M500 in-car video systems and view the captured footage. This document will walk you through configuring the M500 and connecting it to your instance of VideoManager.

This document assumes you have already installed the M500 vehicle hardware, and are familiar with its usage. For more information about doing so, please see the [installation guide](#) and [user guide](#).

Create a group for use with the M500

Unlike body-worn cameras, where one user is assigned to one camera, the M500 should be assigned to a group of users, all users within this group will then be able to log into using the control panel,, and operate the M500 in the field. You must manually create this group on VideoManager. To do so:

1. Navigate to the **Admin** tab.
2. Select the **People** pane.
3. Click the **Groups** section.
4. Click **Create Group** in the top right-hand corner.
5. Enter a name for the group. (for example 'M500 Users')
6. In the right-hand **Roles** pane, ensure that **Vehicle Operator** is set to **On** for the group.



Roles	
System administrator	Off
Device operator	Off
Media importer	Off
Media manager	Off
Media supervisor	Off
System manager	Off
System supervisor	Off
System user	Off
Vehicle Operator	On

7. Click **Create group**.

You should then add all of the users who will be operating M500s to this group. To do so:

1. Navigate to the **Admin** tab.

2. Select the **People** pane.
3. Click the **Users** section.
4. Click **Go to user** next to the relevant user.
5. In the **Group memberships** section, enter the previously-created group name and click **+**.
6. Click **Save user**.

Configure media fields for the M500

Media fields dictate how videos are categorised on the M500 once recording has stopped. In addition to a mandatory event category field, you can also optionally add up to five other media fields to the M500. This document will cover both steps.

Edit the M500 event category media field

VideoManager comes with a built-in event category field, which enables users to categorise the videos they capture while still in the field on the M500 itself. To edit this field to meet your needs:

1. Navigate to the **Admin** tab.
2. Select the **Policies** pane.
3. Click the **User-Defined Media Fields** section.
4. Click **Edit field** next to the *EventCategory* field.

Media type	media_type	Text	No	>
Status	media_status	Text	No	>
Verification	media_verification	Text	No	>
Scheduled deletion	media_scheduled_deletion	Text	No	>
EventCategory	EventCategory	Drop down	No	>

5. In the **Display name** field, enter the name which users will see when they are prompted to complete this field.
6. If **Mandatory** is set to **Yes**, users must complete this field before they can proceed.
7. From the **Display timeout** dropdown, select how long this field will stay on the M500's screen before it disappears.

This is only an option if the field is not mandatory.

8. From the **Permission group** dropdown, select which access group will be able to see this field when viewing the video on VideoManager.
9. From the **Column width** dropdown, select how much space this field will take up when displayed on VideoManager.

Finally, scroll down to the **Drop down** section. VideoManager comes with dropdown values for this field by default - you can choose to edit or delete these values, or create your own. To do so:

1. In the **Values** table, you can configure settings relating to the field's values:
 - a. To **create** a new value, which users can choose to categorise their video with, click **+** in the top right-hand corner. You can then configure:
 - i. The name of the value.
 - ii. The display name of the value.
 - iii. Whether the value is **critical** (i.e. the recording will be prioritised during upload, its resolution may be affected if you configure critical rules to determine image quality, and an incident will be created containing the recording upon import).
 - b. To **edit** an existing value, click **Edit value**. You can then change its name, display name, and critical status.
 - c. To **delete** an existing value, click **Delete value**.
2. From the **Default value** dropdown, select which value will be selected by default.
3. If **Include in match text search** is set to **Yes**, M500 videos can be filtered by the value selected from this dropdown using the **Match Text** field in the **Search Media** pane.
4. If **Show search field** is set to **On**, a field will appear in the **Search Media** pane that enables users to filter videos **only** by the value selected from this dropdown.
5. If **Show in Incidents** is set to **On**, this dropdown's value will be viewable in the **Incident clips** section of the incident editor once the video has been added to an incident.
6. If **Show in summary** is set to **Yes**, this field will be visible when users edit and view incidents.

Finally, scroll down to the **Conditions** section. You can configure whether this field only appears in a video's **Edit properties** pane when another drop down field or check box field has been populated in a specific manner. To do so:

1. Ensure there is at least **one** drop down field or check box field on VideoManager already. Conditions can only be used on these fields.
2. Click **New condition** in the top right-hand corner.
3. From the **Field** dropdown, select which drop down field or check box field will dictate the appearance of this field.

4. From the **Value** dropdown, select which of the drop down field or check box field's values will dictate the appearance of this field.
 - a. This will be presented as a checkbox if a check box field has been chosen, and a dropdown if a drop down field has been chosen. Click **confirm**.

Click **Save settings** to save the event category field.

Optionally add other M500 event tags

It is possible to make other existing media fields compatible with the M500, so users can categorise the videos they capture while still in the field on the M500 itself. These are called event tags. Up to five media fields can be enabled as event tags, in addition to the event category. To do so:

1. Navigate to the **Admin** tab.
2. Select the **Policies** pane.
3. Click the **User-Defined Media Fields** section.
4. Click **Edit field** next to the relevant field.
5. Set **Is Event Tag?** to **On**.
6. Click **Save settings**.

 If you create a new user-defined media field with the following settings, operators can use the M500 control panel to specify the title of the incident which will be created automatically when a critical recording is saved:

- Set **Identifier** to **title**.
- Set **Is Event Tag?** to **On**.

 For more information about creating user-defined media fields, please see the VideoManager admin guide.

Optionally configure incident fields and rules for the M500

Once you have configured media fields for the M500, you can dictate whether these fields are synchronized to corresponding incident fields (meaning they will be visible in an incident form containing the video), and whether VideoManager will automatically create incidents from videos tagged as critical.

Optionally create an M500 event category incident field

The M500 event category **media** field enables operators to categorise recordings on the M500 itself. However, by default, the value selected in this field will only appear alongside the individual recording - it will *not* appear in incidents the recording is added to. To synchronize this media field, so its value appears in associated incidents, you must create a new **incident** field. To do so:

1. Navigate to the **Admin** tab.
2. Select the **Policies** pane.
3. Click the **User-Defined Incident Fields** section.
4. Click **Create field** in the top right-hand corner.
5. In the **Identifier** field, enter *EventCategory*.

⚠ The identifier must be set to *EventCategory*, or the two fields will not synchronize.

6. Ensure **Mandatory** has been set to **No**.
7. Scroll down to the **Drop down** section. You must recreate all of the *EventCategory* media field's values. To do so, click **+** in the top right-hand corner, and add each value one by one.

⚠ The incident field must have the exact same values as the media field, or the two fields will not synchronize.

8. Click **Save settings**.

Optionally enable and configure automatic incident creation

You can configure VideoManager to automatically create an incident for each video which has been tagged as critical on the M500, as soon as those videos have been imported. To do so:

1. Navigate to the **Admin** tab.
2. Select the **Policies** pane.
3. Click the **Auto Incident Creation** section.
4. Set **Auto incident creation enabled** to **On**.
5. Set **Use whole recording** to **On**.
6. In the **Auto incident creation criteria** field, enter:

```
isCritical()
```

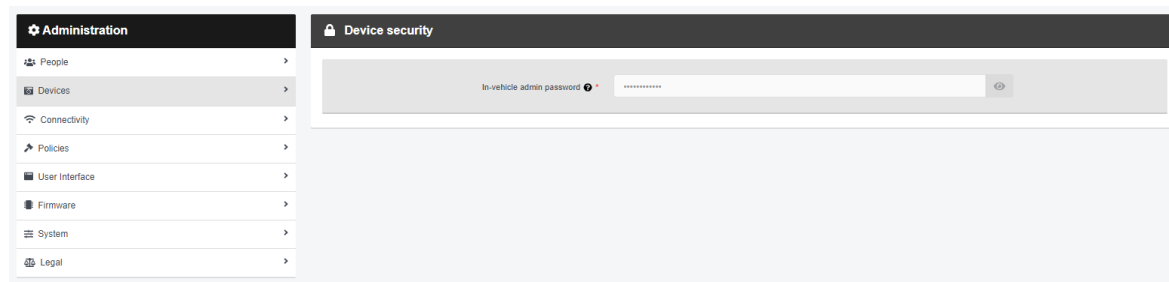
7. Click **Save settings**.

Set a in-vehicle administrator password

Only users who know the in-vehicle administrator password can change administrative settings on their M500.

To set this password:

1. Navigate to the **Admin** tab.
2. Select the **Devices** pane.
3. Click the **Device Security** section.



4. In the **In-vehicle admin password** field, enter the password and make a note of it somewhere secure.
5. Click **Save settings**.

Create a vehicle network profile

You must create a vehicle network profile for your M500(s). This will enable it to connect to the router in the car, and by extension to VideoManager. To do so:

1. Navigate to the **Admin** tab.
2. Select the **Connectivity** pane.
3. Click the **Vehicle network profiles** section.

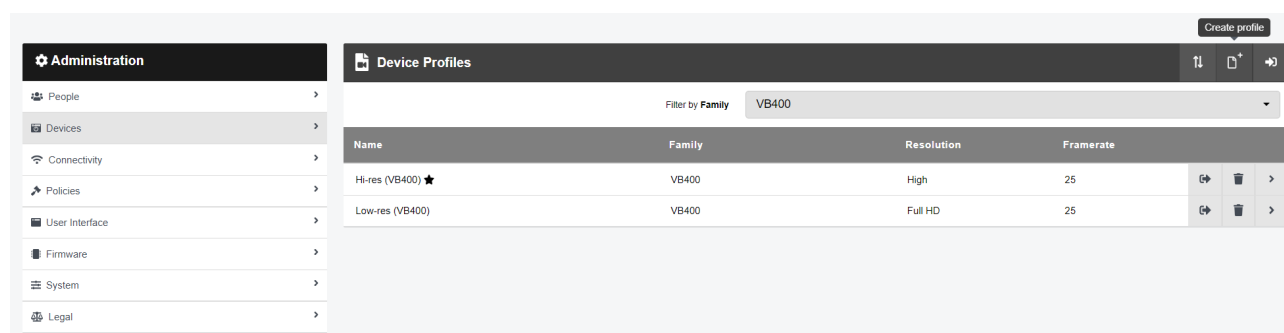
 A screenshot of the 'Create Vehicle network profile' form. The form has a 'Back' button at the top left. It is divided into three main sections: 'Details', 'Wireless radio connection', and 'Local area network'.
 - In the 'Details' section, there is a 'Name' field with a placeholder 'Enter name', a 'Default profile' dropdown set to 'No', and a 'Radio IP address' field set to '10.1.0.1'.
 - In the 'Wireless radio connection' section, there is a 'Wireless type' dropdown set to 'Mikrotik', a 'User name' field set to 'admin', and a 'Password' field.
 - In the 'Local area network' section, there is a 'DHCP client' dropdown set to 'On'.
 A 'Save profile' button is located at the bottom right of the form.

4. In the **Details** section, enter a name for this network profile.
5. From the **Wireless types** dropdown, select the type of router which the M500 will connect to in the car.
If you are unsure, Motorola Solutions recommends choosing **Wired**.
6. Set **DHPC Client** to **On** if the network router should determine the M500's IP, gateway and subnet mask.
Alternatively, set **DHPC Client** to **Off** if you wish to specify the M500's network credentials manually.

Create an M500 device profile

Device profiles dictate every aspect of your M500's behavior in the field. To create a new profile: Every device that is assigned with the same device profile will use the same settings.

1. Navigate to the **Admin** tab.
2. Select the **Devices** pane.
3. Click the **Device Profiles** section.
4. Click **Create profile**.



5. In the **Name** field, enter a name for the device profile.
6. From the **Device** family dropdown, select **M500**.

General tab

- From the **Time Zone** dropdown, specify the time zone in which your M500s are operating.
- From the **Temperature units** dropdown, specify whether the ambient temperature of the car is reported in **Fahrenheit** or **Celcius**.
- From the **Distance units** dropdown, specify whether the distance covered by the car is reported in **Kilometers** or **Miles**.

Devices tab

The following instructions are broken down by section:

On-Screen text management

Select which information will be presented to the M500 user as text on its screen.

GPS section

- From the **Format** dropdown, select how GPS coordinates are formatted on the M500 screen.

 This option will only take effect if **GPS Coordinates** has been enabled from the **On-Screen text management** section.

- If **Include vehicle speed in metadata file** is set to **On**, speed information will be stored in the metadata file of videos recorded on M500s with this device profile.

In-Car officer permissions

- If **Allow modification of date/time** is set to **On**, users will be able to manually change the date and time of the M500.
- From the **Review video** dropdown, select which videos can be reviewed on the M500 by users:
 - **No video review access** - users cannot review videos on the M500. They still may be able to do so on VideoManager, depending on their permissions,
 - **Only video since last login** - users can only review videos on the M500 which were recorded since they last logged in.
 - **Access to all videos** - users can review all videos stored on the M500's HDD (normally around 4-5 days).

Power behavior

- If **Power DVR ON automatically with ignition** is set to **On**, the M500 will turn on automatically when the vehicle ignition is started.
- If **Power DVR OFF automatically with ignition** is set to **On**, the M500 will turn off automatically when the vehicle ignition is turned off - if a time has been set from the **Ignition shutdown timer field**, then the M500 will shut down after this timer expires.
- If **Allow DVR to power OFF while ignition is On** is set to **On**, the M500 can be turned off even if the vehicle's ignition is on.
- If **Stop recording when Ignition Shutdown Timer expires** is set to **On**, the M500 will stop recording when the vehicle's ignition is stopped, and the shutdown timer has expired.

- In the **Ignition shutdown timer** field, enter the number of minutes for which the M500 will stay on after the ignition has been turned off.
- In the **Wireless transfer shutdown timer** field, enter the number of minutes for which the M500 will continue to wirelessly offload footage to VideoManager **after** the ignition has been turned off. The value entered here is added to the time already configured from the **Ignition shutdown timer** field.

 If there is no footage to offload, the M500 will power down immediately.

Other

- If **Blank screen on movement** is set to **On**, the M500's screen will immediately show a static M500 splash screen when it detects the car is moving. (No moving images will be shown)
- If **Allow covert mode** is set to **On**, the M500 will enter covert mode when its **On** button is held down for a few seconds. In this mode, the screen and button LEDs will go off, and it will not make any noise.

Recording tab

Camera

- If **Enable Record-After-the-Fact** is set to **On**, the M500 will continuously record and save video in a continuous loop until the hard drive is full, after which, it will record over the oldest footage. (It will not record over event recordings that have not yet been uploaded)
- If **Use critical rules in-car to determine quality** is set to **On**:
 - Videos tagged with a critical event category will be saved in the configured primary stream resolution.
 - Videos tagged with a non-critical event category will be saved in the configured secondary stream resolution.
- If **Allow increased video quality** is set to **On**, the user can manually upgrade the resolution of a video tagged with a non-critical event category.

Watermarks

- If **Motorola Solutions logo** is set to **On**, the Motorola Solutions logo will appear over every frame of a video recorded on the M500.
- If **Time stamp watermark** is set to **On**, a time stamp will appear over every frame of a video recorded on the M500.
- If **Device identifier watermark** is set to **On**, the M500's serial number will appear over every frame of a video recorded on the M500.

Front camera, Front camera wide, Cabin camera, and Rear camera

Configure the resolution, quality and frame rate for each camera and corresponding stream.

 Selecting a higher video quality results in larger storage space needs. The following table lists the supported video quality options for the M500 system camera views with their storage space needs:

Video quality	Video resolution	Frames per second	Storage space (GB per hour)
HD high resolution	1920 x 1080	30	4.50
		15	2.25
		10	1.50
		5	0.75
	1280 x 720	30	2.25
		15	1.125
		10	0.75
		5	0.375
HD medium resolution	1920 x 1080	30	3.60
		15	1.80
		10	1.20
		5	0.60
	1280 x 720	30	1.80
		15	0.90
		10	0.60
		5	0.30
HD low resolution	1920 x 1080	30	2.70
		15	1.35
		10	0.90
		5	0.45
	1280 x 720	30	1.35
		15	0.675
		10	0.45
		5	0.225
SD high resolution	864 x 480	30	0.90

		15	0.45
		10	0.30
		5	0.15
SD medium resolution	864 x 480	30	0.675
		15	0.338
		10	0.225
		5	0.113
SD low resolution	864 x 480	30	0.45
		15	0.225
		10	0.15
		5	0.075

You can also configure whether users can turn off the front camera wide, cabin camera or rear camera, using the **User permissions** dropdown. Options are as follows:

- **Force camera on** - users cannot change the camera state from the M500. The camera remains on if the M500 system is on. Users are not allowed to turn off the camera.
- **Camera on or off** - officers are allowed to turn off the camera from the control panel.

 The front camera is always on and cannot be turned off.

Wired 1, Wired 2 and cabin camera microphone

- If **Allow muting** is set to **On**, users can turn off the relevant microphone regardless of recording status.
- From the **Recording** dropdown, select when the relevant microphone will capture audio. The options are as follows:
 - **Always on**
 - **During event recording**
 - **Never**
- If **Unmute at start of recording** is set to **On**, the relevant microphone will unmute itself upon recording, even if the user has manually turned it off.

Recording reminder alerts

- If **Active recording** is set to **On**, you can configure how often the M500 will alert the user to the fact that an event is being recording (via a beep). The intention is to prompt and officer to stop the event recording when appropriate.

Additional record time

- From the **Pre-event time** dropdown, select how much footage the M500 will retain from the period before recording was initiated.

Auto-start record

Configure which in-car events will trigger the M500 to start recording automatically

- If **Patrol speed** is enabled, you can configure the speed in MPH / KPH which triggers recording.
- If **Use delay** is enabled for Lights/Sirens/Auxiliary 1/Auxiliary 2, you can select from the **Delay** dropdown how much of a delay there is between the event and the M500 initiating recording.

Auto-stop record

- From the **Prompt for stop** dropdown, select how long the M500 will wait after recording is automatically started, to prompt the user to stop recording.
- If **Maximum recorded event time** is set to **On**, you can configure the maximum length of the video recorded on an M500, before recording will be stopped automatically.

USB groups

In the **Rewrite after** field, you can specify the period of time after which the M500 will start re-writing footage to a USB stick.

 Normally, when a piece of M500 footage is exported to a USB stick, the M500 “marks” the footage so it will not be exported again. However, after the time period specified in this field, the M500 “unmarks” the footage and will re-export it to a USB stick, if the USB stick is inserted.

Event tags

In the **Assigned event tags** table, optionally enable the previously-created event tags.

Network tab

Upload server

By default, your M500 will connect to VideoManager using VideoManager’s public address. If it should connect to VideoManager via another address, set **Use custom upload server** to **On**.

DNS server

Use custom DNS server should only be set to **On** if you are uploading M500 footage to an on-premises instance of VideoManager. If this is the case, enter the URL or IP address of your DNS server.

Click **Save settings**.

Add your M500 to VideoManager

You can now add your M500 to VideoManager. To do so:

1. Navigate to the **Devices** tab.
2. Select the **Search Vehicles** pane.
3. Click **Advanced** in the top right-hand corner, and select **Configure new M500**.

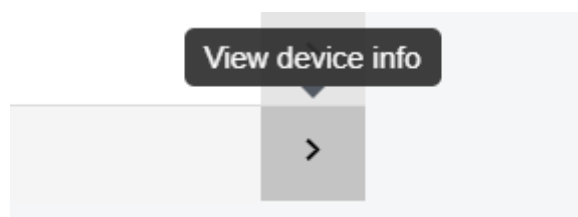
4. In the **Vehicle name** field, enter the name of the vehicle. This is normally the vehicle's number plate.
5. In the **Operator name** field, enter the name of the previously-created **group**.
6. From the **Device profile** dropdown, select the previously-created M500 device profile.
7. From the **Vehicle network profile** dropdown, select the previously-created M500 network profile.
8. Click **Create M500**.

The M500 will appear in the **Search Vehicles** tab - however, the DVR must still be configured before it can be used. To do so:

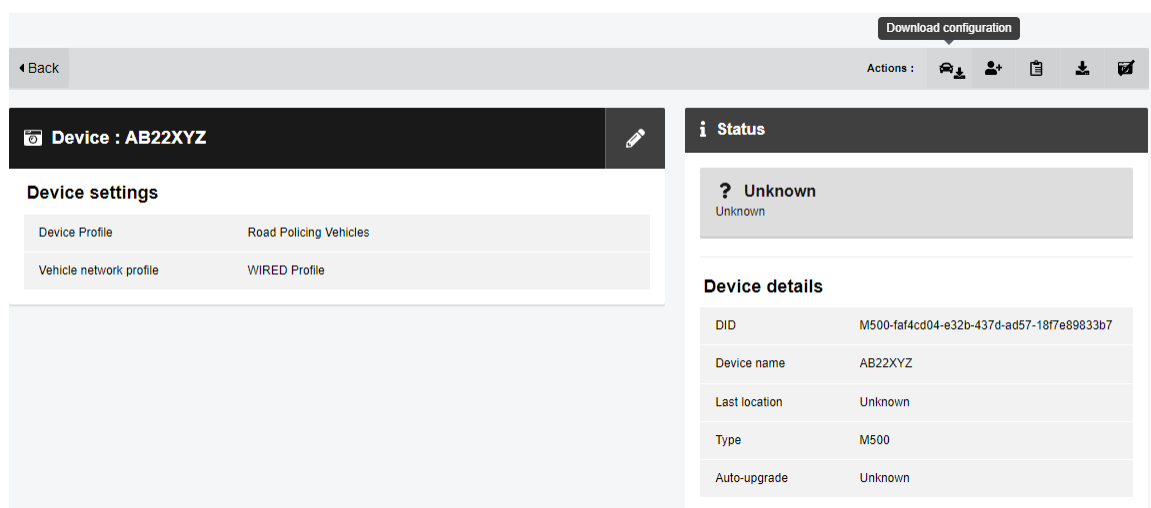
1. Click **Assign Device**.

Device	Operator	Status	
? AB22XYZ	vehicle-operators (Vehicle operators)	Unknown	 

2. Enter the same information as in the previous step, and click **Assign Device**.
3. Click **View device info**.



4. Click **Download configuration**.



5. Drag and drop the downloaded configuration file to a FAT32-formatted USB stick, **in the following folder**:

\Configurations\

6. Plug the USB stick into the M500.

If your M500 is new, it will show you all of the individual XML config files on the memory stick. Choose the correct config file that is appropriate to the M500 you are configuring and press **Load**. The M500 will then reboot and adopt the device profile.

If your M500 has been configured already, it may ask you for an administrative password to load the new config.

Move an M500 to a new vehicle

If you need to move your M500 to a new vehicle, you must clear the M500's serial number, then create a new vehicle configuration for it. The old configuration should **not** be deleted, as it will be linked to historic footage.

To clear an M500's serial number:

1. Navigate to the **Devices** tab.
2. Select the **Search Vehicles** pane.
3. Click **View device info** next to the M500 which will be moved.
4. Click **Clear serial number** in the top right-hand corner.

You can now create a new vehicle profile and apply it to the M500. Please see [this section](#) for more information on how to do so.

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■ The European Union's WEEE directive requires that products sold into EU countries must have the crossed out trash bin label on the product (or the package in some cases).

As defined by the WEEE directive, this cross-out trash bin label means that customers and end-users in EU countries should not dispose of electronic and electrical equipment or accessories in household Waste.

Customers or end-users in EU countries should contact their local equipment supplier representative or service centre for information about the waste collection system in their country.

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